REDUCTION MODULO p AND A BOUND ON THE TORSION: SOLUTIONS

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The aim is a `Magma` script, written from scratch, that bounds the torsion subgroup of an elliptic curve over $\mathbb{Q}$ using only the local theory of [Sil09, §§VII.1–VII.3]. Section 1 shows that reduction modulo an odd prime of good reduction is injective on the rational torsion points, section 2 turns this into two bounds, and section 3 tests them against the LMFDB. Each problem prints a piece of the script with its output. You prove the mathematics behind it. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by a Weierstrass equation with coefficients $a_1, \dots, a_6 \in \mathbb{Z}$ and discriminant Δ.
- The torsion subgroup $T \coloneqq E(\mathbb{Q})_{\mathrm{tors}}$.
- A prime p is good if $p \nmid \Delta$. Then reducing the equation modulo p gives an elliptic curve E_p over $\mathbb{F}_p$.
- For an abelian group G, $G[m]$ is the subgroup killed by m.
- A finite abelian group G is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$ for at most one pair of integers $1 \leqslant n_1 \mid n_2$. When it is, we write $n_1(G)$ and $n_2(G)$ for them.

How to use this. Allow 20 minutes for §1 and 30 for §2. Section 3 is for home. Every `Magma` block runs in seconds in the free online calculator. Paste them in order, since later blocks use earlier functions.

1. Reduction is injective on torsion

Everything in this section is in [Sil09, §§VII.1–VII.3]. The curve of Problem 1.6 is from Application VII.3.5 there.

In the following definition and results, p is a good prime for E.

Definition 1.1 (Reduction modulo p)**.** We regard $E(\mathbb{Q})$ as a subgroup of $E(\mathbb{Q}_p)$, and write $P \mapsto \bar{P}$ for the reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ and $E_1(\mathbb{Q}_p)$ for its kernel.

Proposition 1.2 (Minimality [Sil09, Remark VII.1.1])**.** *The equation is minimal at p, because $v_p(\Delta) = 0 < 12$. So E_p is the reduction of E modulo p.*

Proposition 1.3 (Reduction is a homomorphism [Sil09, Proposition VII.2.1 and the discussion after it])**.** *The reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ is a surjective group homomorphism. Its kernel $E_1(\mathbb{Q}_p)$ consists of O and the $(x, y) \in E(\mathbb{Q}_p)$ with $v_p(x) < 0$.*

Proposition 1.4 (Torsion prime to p [Sil09, Proposition VII.3.1])**.** *If $p \nmid m$, then $E_1(\mathbb{Q}_p)[m] = \{O\}$, and the reduction map above restricts to an injection $E(\mathbb{Q}_p)[m] \hookrightarrow E_p(\mathbb{F}_p)$.*

Theorem 1.5 (Torsion of p-power order [Sil09, Theorem VII.3.4(b)])**.** *Let $P \in E(\mathbb{Q}_p)$ have exact order p^n with $n \geqslant 1$, and let $r \coloneqq \lfloor 1/(p^n - p^{n-1}) \rfloor$. Then $p^{2r}x(P) \in \mathbb{Z}_p$. In particular $x(P) \in \mathbb{Z}_p$ unless $p = 2$ and $n = 1$.*

Date: October 14, 2026.

Problem 1.6 (Injectivity of reduction on torsion)**.**

(1) Let $p \geqslant 3$ be a good prime. Show that the reduction map

$$E(\mathbb{Q}) \hookrightarrow E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$$

is injective on T. In particular T is finite, and isomorphic to a subgroup of $E_p(\mathbb{F}_p)$.

Solution. By Proposition 1.3 it suffices to show that $T \cap E_1(\mathbb{Q}_p) = \{O\}$. Suppose $P \in T \cap E_1(\mathbb{Q}_p)$ has order $m \geqslant 2$, let ℓ be a prime dividing m, and put $Q := (m/\ell)P$, which lies in $E_1(\mathbb{Q}_p)$ and has order ℓ. If $\ell \neq p$, this contradicts Proposition 1.4. If $\ell = p$, then Theorem 1.5 with $n = 1$ gives $x(Q) \in \mathbb{Z}_p$, since $p \geqslant 3$, whereas $v_p(x(Q)) < 0$ by Proposition 1.3. This is a contradiction. So reduction is injective on T, and T embeds in the finite group $E_p(\mathbb{F}_p)$.

Remark. No global input, such as the Mordell–Weil theorem, was used.

(2) The prime 2 is different: even if $p = 2$ is good, $T \to E_2(\mathbb{F}_2)$ need not be injective. Take for instance the curve $E\colon y^2 + xy = x^3 + 4x + 1$:

```
E := EllipticCurve([1, 0, 0, 4, 1]);            // y^2 + xy = x^3 + 4x + 1
Factorization(Integers()!Discriminant(E));      // [ <5, 2>, <13, 2> ]
P := E![-1/4, 1/8];  2*P eq E!0;                // true
```

So 2 is a good prime and $P = (-1/4, 1/8)$ is a rational point of order 2. Show that $\bar{P} = O$ in $E_2(\mathbb{F}_2)$. Which hypothesis of Proposition 1.4 and Theorem 1.5 fails, and what does Proposition 1.4 still give at $p = 2$?

Solution. **The hypothesis $p \nmid m$ fails. Torsion of odd order still injects.** Clearing denominators, $P = [-2 : 1 : 8]$, which reduces to $[0 : 1 : 0] = O$ modulo 2. So the distinct torsion points O and P have the same reduction. Proposition 1.4 needs $p \nmid m$, but $p = m = 2$. For $p = 2$ and $n = 1$, Theorem 1.5 has $r = 1$ and only gives $4x(P) \in \mathbb{Z}_2$ (which is true for $x(P) = -1/4$). Proposition 1.4 still shows that the torsion points of odd order inject into $E_2(\mathbb{F}_2)$.

Remark. The script skips $p = 2$.

2. Two bounds

This section is [Sil09, Application VII.3.2], and its three curves are [Sil09, Examples VII.3.3.1–VII.3.3.3].

Problem 2.1 (The order bound)**.**

The first piece of the script computes the greatest common divisor b of the orders $\#E_p(\mathbb{F}_p)$ over the good odd primes $p \leqslant N$:

```
// b = gcd of #E_p(F_p) over the good odd primes p <= N (there must be one)
OrderBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  return GCD([#ChangeRing(E, GF(p)) : p in good]);
end function;
E1 := EllipticCurve([0, 0, 1, -1, 1]);          // y^2 + y = x^3 - x + 1
Discriminant(E1);                               // -611
[#ChangeRing(E1, GF(p)) : p in [3, 5, 7]];      // [ 1, 8, 6 ]
OrderBound(E1, 30);                             // 1
```

```
E2 := EllipticCurve([0, 3]);                          // y^2 = x^3 + 3
Discriminant(E2);                                     // -3888
[#ChangeRing(E2, GF(p)) : p in [5, 7, 11, 13]];       // [ 6, 13, 12, 9 ]
OrderBound(E2, 30);                                   // 1
```

(1) Deduce that both curves have trivial torsion, and that $y^2 = x^3 + 3$ has infinitely many rational points.

Solution. By Problem 1.6(1), $\#T$ divides $\#E_p(\mathbb{F}_p)$ for every good odd prime p, hence divides b. Both outputs are $b = 1$: for $y^2 + y = x^3 - x + 1$ already $\#E_3(\mathbb{F}_3) = 1$, and for $y^2 = x^3 + 3$ the orders 6 and 13 at $p = 5, 7$ are coprime. So $T = \{O\}$ for both curves. The rational point $(1, 2)$ is not O, so it has infinite order, and its multiples are infinitely many distinct rational points.

(2) `OrderBound` skips the primes dividing the discriminant of the equation it is given. Compare two equations for the first curve, with $N = 7$:

```
E4 := EllipticCurve([0, 0, 27, -81, 729]);      // y^2 + 27y = x^3 - 81x + 729
Discriminant(E1), Discriminant(E4);             // -611 -324710451
IsIsomorphic(E1, E4);                           // true
OrderBound(E1, 7), OrderBound(E4, 7);           // 1 2
```

Here $-324710451 = -3^{12} \cdot 611$. Find a change of variables between the two equations. Explain why a non-minimal equation can make the script skip a prime it could have used, but never makes it use one it should not.

Solution. $\boldsymbol{x = 9x', y = 27y'}$. Substituting $x = 9x'$, $y = 27y'$ in $y^2 + 27y = x^3 - 81x + 729$ gives

$$3^6(y'^2 + y') = 3^6(x'^3 - x' + 1), \quad \text{that is,} \quad y'^2 + y' = x'^3 - x' + 1.$$

This change of variables has $u = 3$, and the two discriminants differ by $u^{12} = 3^{12}$ [Sil09, §III.1]. So 3 is a good prime for $y^2 + y = x^3 - x + 1$, although it divides the discriminant of the other equation. For any equation with integer coefficients, a prime not dividing its discriminant is good, and Proposition 1.2 makes the equation minimal there, so Problem 1.6(1) applies. A non-minimal equation only has more primes dividing its discriminant, which the script skips: the bound can get weaker, never wrong. Here, with $N = 7$, the minimal equation uses $p = 3, 5, 7$ and $\#E_3(\mathbb{F}_3) = 1$ gives $b = 1$, while the other equation only uses $p = 5, 7$ and gives $b = \gcd(8, 6) = 2$.

Corollary 2.2 (The m-torsion [Sil09, Corollary III.6.4]). *Let E be an elliptic curve over a field K, and $m \geqslant 1$. If $\operatorname{char} K \nmid m$, then $E[m](\overline{K}) \cong (\mathbb{Z}/m\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$, then $E[p^e](\overline{K})$ is cyclic for every $e \geqslant 1$.*

Problem 2.3 (The structure bound).
The curve $y^2 = x^3 + x$ shows that the orders are not the whole story. `Invariants` returns $[n_1(G), n_2(G)]$ for $G = E_p(\mathbb{F}_p)$, omitting $n_1(G)$ when it is 1, and returns [] when G is trivial:

```
E3 := EllipticCurve([1, 0]);                          // y^2 = x^3 + x
Discriminant(E3);                                     // -64
OrderBound(E3, 30);                                   // 4
[Invariants(AbelianGroup(ChangeRing(E3, GF(p)))) : p in [3, 5, 7, 11, 13]];
// [ [ 4 ], [ 2, 2 ], [ 8 ], [ 12 ], [ 2, 10 ] ]
```

(1) Let G be a finite subgroup of $E(\overline{K})$, for an elliptic curve E over a field K. Show that $n_1(G)$ and $n_2(G)$ are defined, and that $n_2(G)$ is the exponent of G. Show that $\#E[n_1(G)](\overline{K}) = n_1(G)^2$, and that an integer $m \geqslant 1$ with $\#E[m](\overline{K}) = m^2$ satisfies $E[m](\overline{K}) \subseteq G$ if and only if $m \mid n_1(G)$. Deduce that $n_1(H) \mid n_1(G)$ and $n_2(H) \mid n_2(G)$ for every subgroup $H \subseteq G$.

SOLUTION. By the structure theorem,

$$G \cong \mathbb{Z}/d_1\mathbb{Z} \times \cdots \times \mathbb{Z}/d_k\mathbb{Z}, \qquad 1 \leqslant d_1 \mid \cdots \mid d_k,$$

for some integers $d_1, \ldots, d_k$ with $k \geqslant 2$. Take k as small as possible. If $k > 2$, then $d_1 > 1$, since otherwise the factor $\mathbb{Z}/d_1\mathbb{Z}$ could be dropped. For a prime $\ell \mid d_1$, the subgroup $G[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^k$ lies in $E[\ell](\overline{K})$, which has at most ℓ^2 elements by Corollary 2.2, a contradiction. So $k = 2$, and $n_1(G) = d_1$, $n_2(G) = d_2$ are defined. The exponent of $\mathbb{Z}/n_1(G)\mathbb{Z} \times \mathbb{Z}/n_2(G)\mathbb{Z}$ is $n_2(G)$.
Put $n_1 := n_1(G)$, so that $G[n_1] \cong (\mathbb{Z}/n_1\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$ and $p \mid n_1$, then $G[p] \cong (\mathbb{Z}/p\mathbb{Z})^2$ would lie in the cyclic group $E[p](\overline{K})$, which is impossible. So $\operatorname{char} K \nmid n_1$, and $E[n_1](\overline{K}) \cong (\mathbb{Z}/n_1\mathbb{Z})^2$ has n_1^2 elements. It contains $G[n_1]$, which has as many, so $E[n_1](\overline{K}) = G[n_1] \subseteq G$. Hence $m \mid n_1$ implies $E[m](\overline{K}) \subseteq E[n_1](\overline{K}) \subseteq G$. Conversely, if $\#E[m](\overline{K}) = m^2$ and $E[m](\overline{K}) \subseteq G$, then $E[m](\overline{K}) = G[m]$, and

$$m^2 = \#G[m] = \gcd(m, n_1)\gcd(m, n_2(G)) \leqslant \gcd(m, n_1)\, m.$$

So $\gcd(m, n_1) = m$, that is, $m \mid n_1$.
Finally, let $H \subseteq G$. Then $E[n_1(H)](\overline{K}) \subseteq H \subseteq G$ has $n_1(H)^2$ elements, so $n_1(H) \mid n_1(G)$. And $n_2(G)$ kills H, so $n_2(H) \mid n_2(G)$.

(2) Let b_1 and b_2 be the greatest common divisors of $n_1(E_p(\mathbb{F}_p))$ and of $n_2(E_p(\mathbb{F}_p))$ over the good odd primes $p \leqslant N$. Show that $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $b_1 b_2 \mid b$. This is the second piece of the script:

```
// [b1, b2]: n1(T) | b1 and n2(T) | b2
StructureBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  b1 := 0;  b2 := 0;
  for p in good do
    I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
    b1 := GCD(b1, #I eq 2 select I[1] else 1);
    b2 := GCD(b2, #I eq 0 select 1 else I[#I]);
  end for;
  return [b1, b2];
end function;
StructureBound(E3, 30);                          // [ 1, 2 ]
P := E3![0, 0];  2*P eq E3!0;                    // true
```

Conclude that $T = \{O, (0,0)\} \cong \mathbb{Z}/2\mathbb{Z}$ for $y^2 = x^3 + x$. Why does b stay at 4 however large N is?

SOLUTION. **Because $4 \mid \#E_p(\mathbb{F}_p)$ for every prime $p \geqslant 3$.** By Problem 1.6(1), T is isomorphic to a subgroup of $E_p(\mathbb{F}_p) \subseteq E_p(\overline{\mathbb{F}}_p)$ for each good odd prime p, so part (1) gives $n_i(T) \mid n_i(E_p(\mathbb{F}_p))$ for $i = 1, 2$, so $n_i(T) \mid b_i$. Also $b_1 b_2$ divides $n_1(E_p(\mathbb{F}_p))\, n_2(E_p(\mathbb{F}_p)) = \#E_p(\mathbb{F}_p)$ for each p, hence divides b: the structure bound is never worse than the order bound. For $y^2 = x^3 + x$ the output $[b_1, b_2] = [1, 2]$ makes T cyclic of order dividing 2, and $(0,0)$ has order

2. The order bound stays at 4 because $4 \mid \#E_p(\mathbb{F}_p)$ for every prime $p \geqslant 3$ [Sil09, Exercise 5.12]: the orders cannot tell $\mathbb{Z}/4\mathbb{Z}$ from $(\mathbb{Z}/2\mathbb{Z})^2$, but the group structures can.

Remark. Already $p = 3$ and $p = 5$ suffice, with $E_3(\mathbb{F}_3) \cong \mathbb{Z}/4\mathbb{Z}$ and $E_5(\mathbb{F}_5) \cong (\mathbb{Z}/2\mathbb{Z})^2$. This is Silverman's argument.

(3) Show that a rational point P is a torsion point if and only if $b_2P = O$, and that rational torsion points generating a subgroup of order b_1b_2 generate T.

Solution. If $P \in T$, its order divides the exponent $n_2(T)$, which divides b_2. Conversely, $b_2P = O$ says that P has finite order. If rational torsion points generate $H \subseteq T$ with $\#H = b_1b_2$, then $\#T = n_1(T)n_2(T) \leqslant b_1b_2 = \#H$, so $H = T$.

3. How good are the bounds? The LMFDB

The L-functions and modular forms database [LMF25] lists millions of elliptic curves over $\mathbb{Q}$ with their torsion subgroups, computed by methods beyond [Sil09, Chapter VII]. We use it as the answer key.

Proposition 3.1 (Reduction of isogenies [Mil20, proof of Theorem V.4.1])**.** *An isogeny over $\mathbb{Q}$ reduces to an isogeny over $\mathbb{F}_p$ at every good prime p of both equations.*

Proposition 3.2 (Isogenous curves over $\mathbb{F}_p$ [Sil09, Exercise 5.4(a)])**.** *Isogenous curves over $\mathbb{F}_p$ have the same number of rational points.*

Problem 3.3 (Testing the bounds)**.**

(1) Here are the curves of smallest conductor with each torsion structure that occurs over $\mathbb{Q}$, with their LMFDB labels, a-invariants and torsion structures ([1] for the trivial group):

```
curves := [<"11.a1", [0, -1, 1, -7820, -263580], [1]>,
  <"14.a2", [1, 0, 1, -171, -874], [2]>,
  <"19.a2", [0, 1, 1, -9, -15], [3]>, <"15.a4", [1, 1, 1, -80, 242], [4]>,
  <"11.a2", [0, -1, 1, -10, -20], [5]>, <"14.a6", [1, 0, 1, 4, -6], [6]>,
  <"26.b2", [1, -1, 1, -3, 3], [7]>, <"15.a8", [1, 1, 1, 35, -28], [8]>,
  <"54.b2", [1, -1, 1, -14, 29], [9]>, <"66.c3", [1, 0, 0, -45, 81], [10]>,
  <"90.c7", [1, -1, 1, -122, 1721], [12]>, <"15.a2", [1, 1, 1, -135, -660], [2, 2]>,
  <"15.a5", [1, 1, 1, -10, -10], [2, 4]>, <"30.a6", [1, 0, 1, -19, 26], [2, 6]>,
  <"210.e6", [1, 0, 0, -1070, 7812], [2, 8]>];
for c in curves do
  E := EllipticCurve(c[2]);
  printf "%-7o LMFDB %-9o order bound %-3o structure bound %o\n",
    c[1], c[3], OrderBound(E, 50), StructureBound(E, 50);
end for;
// 11.a1   LMFDB [ 1 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a2   LMFDB [ 2 ]     order bound 6   structure bound [ 1, 6 ]
// 19.a2   LMFDB [ 3 ]     order bound 3   structure bound [ 1, 3 ]
// 15.a4   LMFDB [ 4 ]     order bound 8   structure bound [ 1, 4 ]
// 11.a2   LMFDB [ 5 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a6   LMFDB [ 6 ]     order bound 6   structure bound [ 1, 6 ]
// 26.b2   LMFDB [ 7 ]     order bound 7   structure bound [ 1, 7 ]
// 15.a8   LMFDB [ 8 ]     order bound 8   structure bound [ 1, 8 ]
// 54.b2   LMFDB [ 9 ]     order bound 9   structure bound [ 1, 9 ]
// 66.c3   LMFDB [ 10 ]    order bound 10  structure bound [ 1, 10 ]
// 90.c7   LMFDB [ 12 ]    order bound 12  structure bound [ 1, 12 ]
// 15.a2   LMFDB [ 2, 2 ]  order bound 8   structure bound [ 2, 4 ]
```

```
// 15.a5   LMFDB [ 2, 4 ]  order bound 8   structure bound [ 2, 4 ]
// 30.a6   LMFDB [ 2, 6 ]  order bound 12  structure bound [ 2, 6 ]
// 210.e6  LMFDB [ 2, 8 ]  order bound 16  structure bound [ 2, 8 ]
```

Check that no bound contradicts the LMFDB. Where is each bound attained?

SOLUTION. **No contradiction. The bounds are attained eleven and twelve times.** In every row $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $\#T \mid b$. The order bound equals $\#T$ for eleven curves, all but 11.a1, 14.a2, 15.a4 and 15.a2. The structure bound equals $[n_1(T), n_2(T)]$ for twelve, all but 11.a1, 14.a2 and 15.a2.

(2) The curves 15.a4 and 15.a8 are isogenous:

```
E := EllipticCurve([1, 1, 1, -80, 242]);        // 15.a4
F := EllipticCurve([1, 1, 1, 35, -28]);          // 15.a8
good := [p : p in PrimesUpTo(50) | p ne 3 and p ne 5];
[#ChangeRing(E, GF(p)) : p in good] eq [#ChangeRing(F, GF(p)) : p in good];
// true
[<p, Invariants(AbelianGroup(ChangeRing(E, GF(p)))),
     Invariants(AbelianGroup(ChangeRing(F, GF(p))))> : p in [7, 11, 13]];
// [ <7, [ 2, 4 ], [ 8 ]>, <11, [ 2, 8 ], [ 16 ]>, <13, [ 16 ], [ 2, 8 ]> ]
```

Using Propositions 3.1 and 3.2, explain why the order bound cannot separate isogenous curves, and why the structure bound separates these two.

SOLUTION. If E and E' are isogenous over $\mathbb{Q}$, then E_p and E'_p are isogenous over $\mathbb{F}_p$ at each good prime of both equations, so $\#E_p(\mathbb{F}_p) = \#E'_p(\mathbb{F}_p)$ [Sil09, Exercise 5.4(a)], and the order bounds agree. The group structures need not: at $p = 7$ the groups are $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ for 15.a4 and $\mathbb{Z}/8\mathbb{Z}$ for 15.a8, so $b_2 = 4$ for the first and $b_2 = 8$ for the second, and the structure bound is sharp for both.

(3) The LMFDB lists $(5, -2)$ as a torsion generator of 15.a4:

```
E := EllipticCurve([1, 1, 1, -80, 242]);  P := E![5, -2];        // 15.a4
[k*P eq E!0 : k in [1..4]];                       // [ false, false, false, true ]
```

Prove that $T \cong \mathbb{Z}/4\mathbb{Z}$ for 15.a4 without trusting the LMFDB. Can you do the same for 11.a1, 14.a2 or 15.a2?

SOLUTION. **Not for 11.a1, 14.a2 or 15.a2: no choice of primes helps.** The output shows that P has order 4, so $\langle P \rangle$ is a subgroup of T of order $4 = b_1 b_2$, and $T = \langle P \rangle$ by Problem 2.3(3). For 11.a1 the bound $[1, 5]$ allows $T = \{O\}$ or $\mathbb{Z}/5\mathbb{Z}$. To prove $T = \{O\}$ we would have to rule out a rational point of order 5, and the local theory gives no finite list of candidates: it makes the coordinates of a torsion point of order at least 3 integral, and $4x(P)$ integral for a point P of order 2 [Sil09, Application VII.3.5], but it does not bound them. Likewise 14.a2 needs a rational point of order 3 ruled out, and 15.a2 one of order 4. No choice of primes helps. The isogenous curves 11.a2, 14.a6 and 15.a5 have rational torsion of orders 5, 6 and 8, and the same good primes as their partners: the discriminants are -11 and -11^5, $-2^{18} \cdot 7$ and $-2^6 \cdot 7^3$, $3^8 \cdot 5^2$ and $3^4 \cdot 5^4$ (in general, isogenous curves have good reduction at the same primes [Sil09, Corollary VII.7.2]). So by part (2) these orders divide $\#E_p(\mathbb{F}_p)$ at every good odd prime p of 11.a1, 14.a2 and 15.a2, and b never drops below them. Nor does b_2: a prime dividing the order of a finite abelian group divides its exponent, and $8 \mid n_1 n_2$ with $n_1 \mid n_2$ forces $4 \mid n_2$.

Remark. Remark 3.4 says what is missing.

(4) Find these curves yourself: on https://www.lmfdb.org/EllipticCurve/Q/, search by "torsion structure", such as `[2,4]`, and sort by conductor. On a curve's page, find its equation, discriminant, torsion subgroup and bad primes. Pick a curve of your own and compute both bounds.

Solution. The section "Invariants" of a curve's page gives the equation, the a-invariants and the discriminant; "Mordell–Weil group" gives the torsion structure and generators; the table "Local data" lists the bad primes. The LMFDB's torsion must satisfy both bounds, and its generators can be checked as in part (3).

Remark 3.4 (Beyond the local theory)**.** Over $\mathbb{Q}$ the bounds complete to an algorithm. For $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, a rational torsion point $P \neq O$ has integer coordinates, and either $2P = O$ or $y(P)^2 \mid 4A^3 + 27B^2$ (Nagell–Lutz, [Sil09, Corollary VIII.7.2]). This leaves finitely many candidates, and the test $b_2 P = O$ decides which are torsion points. Mazur proved that $E(\mathbb{Q})_{\mathrm{tors}}$ is one of the fifteen groups in the listing [Sil09, Theorem VIII.7.5].

References

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